



SQL Practice Problems

Scenario: Squid Game

Dr. Sumaiya Tabassum Nimi

Assistant Professor, Dept. of ECE — North South University

SQL Problems Using the Squid Game Scenario

Database Schema

Assume the following relational schema:

```
PLAYERS(player_id, name, country, age)
GAMES(game_id, game_name, game_order)
PLAYER_GAMES(player_id, game_id, score, rank_in_game, is_eliminated)
TEAMS(team_id, team_name)
TEAM_MEMBERS(team_id, player_id)
```

Use this schema to answer all questions. The queries should make use of **JOIN**, **GROUP BY**, **HAVING**, and **PARTITION BY** where appropriate.

Part A: Direct Query Writing (Less Complex)

1. Write a query to list each player's name and the game names they have participated in.
2. Write a query to display the player name, game name, and score for all players in the game titled 'Red Light, Green Light'.
3. Write a query to find the total number of players from each country. Display: `country`, `player_count`.
4. (**GROUP BY + HAVING**) Write a query to find all countries that have more than 5 participants. Display: `country`, `player_count`.
5. Write a query to display the average score for each game. Only include games whose average score is at least 70. Display: `game_name`, `average_score`.

Part B: Explain-the-Query (More Complex)

6. Explain in plain English what the following query returns:

```
SELECT p.country, COUNT(DISTINCT p.player_id) AS survivors
FROM PLAYERS p
JOIN PLAYER_GAMES pg ON p.player_id = pg.player_id
JOIN GAMES g ON pg.game_id = g.game_id
WHERE pg.is_eliminated = FALSE
      AND g.game_order = (SELECT MAX(game_order) FROM GAMES)
GROUP BY p.country
HAVING COUNT(DISTINCT p.player_id) >= 2;
```

7. Explain what the following query does and what the column `rank_within_game` represents:

```
SELECT p.player_id, p.name, g.game_name, pg.score,
       RANK() OVER (PARTITION BY g.game_id
                    ORDER BY pg.score DESC) AS rank_within_game
FROM PLAYERS p
JOIN PLAYER_GAMES pg ON p.player_id = pg.player_id
JOIN GAMES g ON pg.game_id = g.game_id;
```

8. Explain step-by-step what the following query produces:

```
WITH total_scores AS (
  SELECT p.player_id, p.name, SUM(pg.score) AS total_score
  FROM PLAYERS p
  JOIN PLAYER_GAMES pg ON p.player_id = pg.player_id
  GROUP BY p.player_id, p.name
),
ranked_by_country AS (
  SELECT t.player_id, t.name, pl.country, t.total_score,
         DENSE_RANK() OVER (
           PARTITION BY pl.country
           ORDER BY t.total_score DESC
         ) AS country_rank
  FROM total_scores t
  JOIN PLAYERS pl ON t.player_id = pl.player_id
)
SELECT country, name, total_score, country_rank
FROM ranked_by_country
WHERE country_rank <= 3
ORDER BY country, country_rank;
```

9. Write a query that shows, for every player: `player_id`, `name`, `country`, and the total number of players from that country as `players_in_country`. (Hint: use `COUNT(*) OVER (PARTITION BY country)`.)
10. Write a query that returns the **top 3 players by score in each game**. Display: `game_name`, `player_name`, `score`, `rank_within_game`. (Hint: use a window function.)